

Hazel Deriver Setup

Step 1

Please share your screen, and open (in Chrome) <https://hazel.org/build/derivation> and this document, each in a separate browser window, with this on the left and tylr on the right. Adjust the width of two windows to about the level shown:

Step 2

Watch the following video introducing Hazel Deriver, the editor you will use to complete a series of small tasks, and the purpose of this study.

VIDEO

Step 3

This is a quick review of what is a derivation, and how to turn a written derivation in to Hazel Deriver one.

VIDEO

Open the link mention in the video.

Task Structure

We will now proceed to the task portion of the study. You will be asked complete 9 small tasks with Hazel Deriver. Each task will be either 1) a transcription task or 2) a modification task or 3) a deriving task. There will be 3 transcription tasks, 3 modification tasks, and 3 deriving tasks. Deriving tasks usually take a bit longer than other tasks, where you will be ask to derive the conclusion with or without infrastructure.

We will time the tasks, though you're encouraged to the complete each task at a comfortable pace. We ask that you attempt to complete each editing task on your own, without help from the study administrator.

After each task, you may be asked to reflect on your editing experience.

Practice Task: Transcription

Setup

1. Switch to the page 1.
2. Turn off Evaluation, Docs, and Highlight.

Goal

Transcribe the following derivation tree you saw into Hazel Deriver. No worry if the derivation is not correct.

let $\mathcal{D}_0$ be

$$\frac{}{\text{False val}} \quad (\text{V-False})$$

in

$$\frac{\frac{}{\text{fun } (x : \text{Unit}) \rightarrow f \ x + \text{unroll}(\underline{5}) \Downarrow \underline{-6}} \quad (?) \quad \mathcal{D}_0}{\vdash ((\), \text{L } \underline{8}).\text{snd} : \text{Num} \times \text{Bool}} \quad (\text{T-Let})$$

Instructions

1. For this practice task, you are encourage to talk to the study administrator about anything about the editing details without turning on **Evaluations** and **Docs**.
2. When you are ready to start, say start. When you are done, say done

Practice Task: Transcription (Cont.)

Compare your answer to the following and check for errors.

```
let d0 =
```

V-False

```
  val False end
```

```
in
```

```
  fun (x : Unit) -> f(x) + unroll(5) \=/ -6 d0
```

T-Let

```
|- (((), L(8)).snd : Num * Bool
```

Task 1 of 9: Transcription

Setup

1. Switch to the page 2.
2. Turn off Evaluation, Docs, and Highlight.

Goal

Transcribe the following derivation tree you saw into Hazel Deriver. No worry if the derivation is not correct.

let $\mathcal{D}_0 =$

$$\frac{\overline{\text{False val}} \quad (\text{V-False})}{\text{False} \Downarrow \text{False}} \quad (\text{E-Val})$$

in

$$\frac{\mathcal{D}_0 \quad \frac{\overline{\text{L False} \Downarrow \text{L False}} \quad (\text{E-InjL}) \quad \frac{\overline{(5, 1) \text{ val}} \quad (\text{V-Pair})}{(5, 1) \Downarrow (5, 1)} \quad (\text{E-Val})}{(5, 1).\text{snd} \Downarrow 1} \quad (\text{E-PrjR})}{\text{if False then L False else } (5, 1).\text{snd} \Downarrow 1} \quad (\text{E-If-F})$$

Instructions

1. Starting from this task, attempt to complete each editing task on your own, without help from the study administrator.
2. The conclusion and prelude variable definition has been provided for you and no revision is needed.
3. When you are ready to start, say start. When you are done, say done

Task 2 of 9: Debugging

Setup

1. Switch to the page 3.
2. Turn on **Evaluation**, keep **Docs**, and **Highlight** off.

Goal

Try to fix the derivation until all the node turn marked correct (turn green).

Instructions

1. Debug the error derivation tree based on the information provided by turning on **Evaluation**.
2. When you are ready to start, say start. When you are done, say done
3. Complete the reflection survey once you are done.

Task 3 of 9: Transcription

Setup

1. Switch to the page 4.
2. Turn on Evaluation, Docs, and turn off Highlight.

Goal

Transcribe the following derivation tree you saw into Hazel Deriver. No worry if the derivation is not correct.

let $\Gamma_a = a : \text{Num}$
 and $\Gamma_{ab} = \Gamma_{ab}, b : \text{Num}$
 and $\mathcal{D}_0 =$

$$\frac{a : \text{Num} \in \Gamma_{ab}}{\Gamma_{ab} \vdash a : \text{Num}} \text{ (T-Var)}$$

and $\mathcal{D}_1 =$

$$\frac{b : \text{Num} \in \Gamma_{ab}}{\Gamma_{ab} \vdash b : \text{Num}} \text{ (T-Var)}$$

in

$$\frac{\frac{\frac{\mathcal{D}_0 \quad \mathcal{D}_1}{\Gamma_{ab} \vdash \text{if } a < b \text{ then } a \text{ else } b : \text{Num}} \text{ (T-Let)}}{\Gamma_a \vdash \text{fun } (b : \text{Num}) \rightarrow \text{if } a < b \text{ then } a \text{ else } b : \text{Num} \rightarrow \text{Num}} \text{ (T-Fun)}}{\vdash \text{fun } (a : \text{Num}) \rightarrow \text{fun } (b : \text{Num}) \rightarrow \text{if } a < b \text{ then } a \text{ else } b : \text{Num} \rightarrow \text{Num} \rightarrow \text{Num}} \text{ (T-Fun)}$$

Instructions

1. The conclusion and prelude variable definition has been provided for you and no revision is needed.
2. When you are ready to start, say start. When you are done, say done

Task 4 of 9: Debugging

Setup

1. Switch to the page 5.
2. Turn on `Evaluation`, and `Docs`. Keep `Highlight` off.

Goal

Try to fix the derivation until all the node turn marked correct (turn green).

Instructions

1. Debug the error derivation tree based on the information provided by turning on `Evaluation`, and `Docs`.
2. The conclusion and prelude variable definition has been provided for you and no revision is needed.
3. When you are ready to start, say start. When you are done, say done
4. Complete the reflection survey once you are done.

Task 5 of 9: Transcription

Setup

1. Switch to the page 6.
2. Turn off Evaluation, Docs, and Highlight.

Goal

Transcribe the following derivation tree you saw into Hazel Deriver. No worry if the derivation is not correct.

let $\Gamma = e : \text{Bool} + \text{Num}$
 and $\Gamma' = \Gamma, y : \text{Num}$
 and $\mathcal{D}_0 =$

$$\frac{\frac{}{\Gamma' \vdash e \Rightarrow \text{Bool} + \text{Num}} \text{ (S-Var)}}{\Gamma' \vdash e \Leftarrow \text{Bool} + \text{Num}} \text{ (A-Sub)}$$

and $\mathcal{D}_1 =$

$$\frac{\frac{\frac{}{\Gamma' \vdash y \Rightarrow \text{Num}} \text{ (S-Var)}}{\Gamma' \vdash y =? 1 \Rightarrow \text{Bool}} \text{ (S-Eq)} \quad \frac{\frac{}{\Gamma' \vdash 1 \Rightarrow \text{Num}} \text{ (S-Num)}}{\Gamma' \vdash y =? 1 \Leftarrow \text{Bool}} \text{ (A-Sub)}}{\Gamma' \vdash L(y =? 1) \Leftarrow \text{Bool} + \text{Num}} \text{ (A-InjL)}$$

in

$$\frac{\mathcal{D}_0 \quad \mathcal{D}_0 \quad \mathcal{D}_1}{\Gamma \vdash \text{case } e \text{ of } L(x) \rightarrow e \text{ else } R(y) \rightarrow L(y =? 1) \Leftarrow \text{Bool} + \text{Num}} \text{ (A-Case)}$$

Instructions

1. The conclusion and prelude variable definition has been provided for you and no revision is needed.
2. When you are ready to start, say start. When you are done, say done

Task 6 of 9: Debugging

Setup

1. Switch to the page 7.
2. Turn on Evaluation, Docs, and Highlight.

Goal

Try to fix the derivation until all the node turn marked correct (turn green).

Instructions

1. Debug the error derivation tree based on the information provided by turning on Evaluation, and Docs.
2. When you are ready to start, say start. When you are done, say done
3. Complete the reflection survey once you are done.

Task 7 of 9: Deriving

Setup

1. Switch to the page 8.
2. Turn on **Evaluation**. Keep **Docs**, and **Highlight** off.

Goal

Try to make a derivation for the following judgement from scratch until all the node turn marked correct (turn green).

let (x, y) be $(2, 2)$ in $x < y \Downarrow \text{False}$

Instructions

1. Write the derivation tree based on the information provided by turning on **Evaluation**. You are encouraged to use the reference table provided for you.
2. When you are ready to start, say start. When you are done, say done
3. Complete the reflection survey once you are done.

Task 8 of 9: Deriving

Setup

1. Switch to the page 9.
2. Turn on **Evaluation** and **Docs**. Keep **Highlight** off.

Goal

Try to make a derivation for the following judgement from scratch until all the node turn marked correct (turn green).

let $\Gamma_x = x : \text{Num}$ in

$$\vdash (\text{fun } (x : \text{Num}) \rightarrow x =? 1)(1) : \text{Bool}$$

Instructions

1. Write the derivation tree based on the information provided by turning on **Evaluation** and **Docs**.
2. When you are ready to start, say start. When you are done, say done
3. Complete the reflection survey once you are done.

Task 9 of 9: Deriving

Setup

1. Switch to the page 10.
2. Turn on Evaluation, Docs, and Highlight.

Goal

Try to make a derivation for the following judgement from scratch until all the node turn marked correct (turn green).

$$\text{let } incr \text{ be fun } x \rightarrow x + \underline{1} \text{ in } incr(\underline{1}) \Downarrow \underline{2}$$

Instructions

1. Write the derivation tree based on the information provided by turning on Evaluation, Docs, and Highlight.
2. When you are ready to start, say start. When you are done, say done
3. Complete the reflection survey once you are done.

End of the Tasks

Thank you for your participation.